

(c) The sum rule states that if g, h are both differentiable at a , then the function $f = g + h$ is also differentiable at a and

$$f'(a) = g'(a) + h'(a).$$

Proof Assuming g, h are differentiable at a . Then.

$$\begin{aligned} & g'(a) + h'(a) \\ &= \left(\lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a} \right) + \left(\lim_{x \rightarrow a} \frac{h(x) - h(a)}{x - a} \right) \\ &= \lim_{x \rightarrow a} \left(\frac{g(x) - g(a)}{x - a} + \frac{h(x) - h(a)}{x - a} \right) \end{aligned}$$

, using the linearity of limiting values.
ie. $\lim A + \lim B = \lim (A + B)$

$$= \lim_{x \rightarrow a} \left(\frac{g(x) - g(a) + h(x) - h(a)}{x - a} \right)$$

, adding the fractions.

